

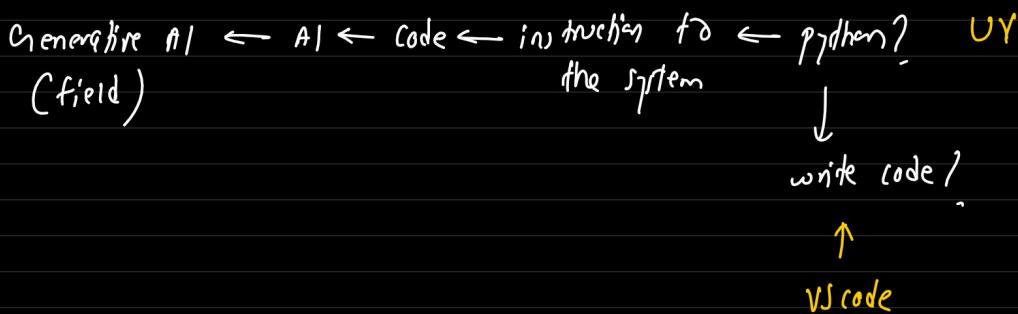
22-03-2026

Agenda:

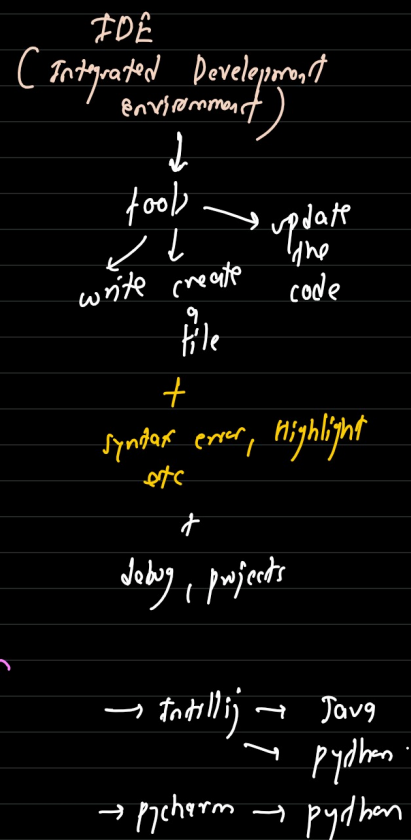
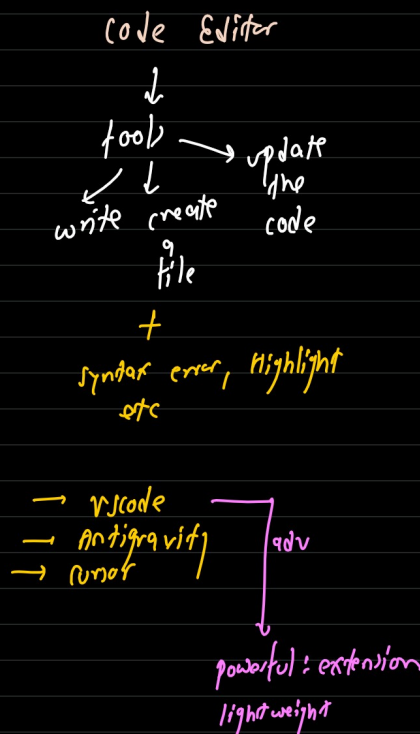
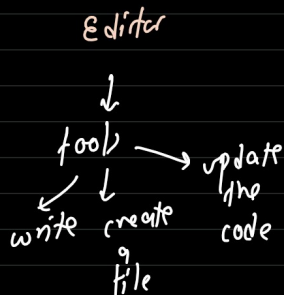
Setup tools

Intro to Generative AI

Tools



Code →



python $\Rightarrow$ programming lang
↓
Generative AI ← Developer + Community

python ← install python → python.org

python ← Anaconda ← Miniconda
↑
more
↑
Anaconda

python ← UV
↑
venv
↑
package manager
↑
python environment
(fast (Rust))
lightweight

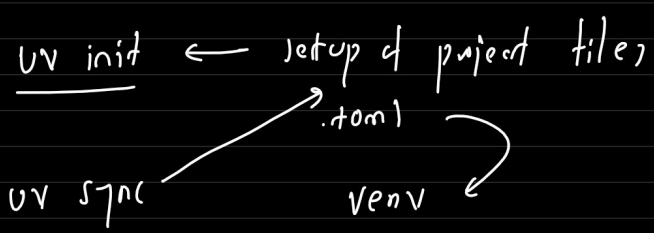
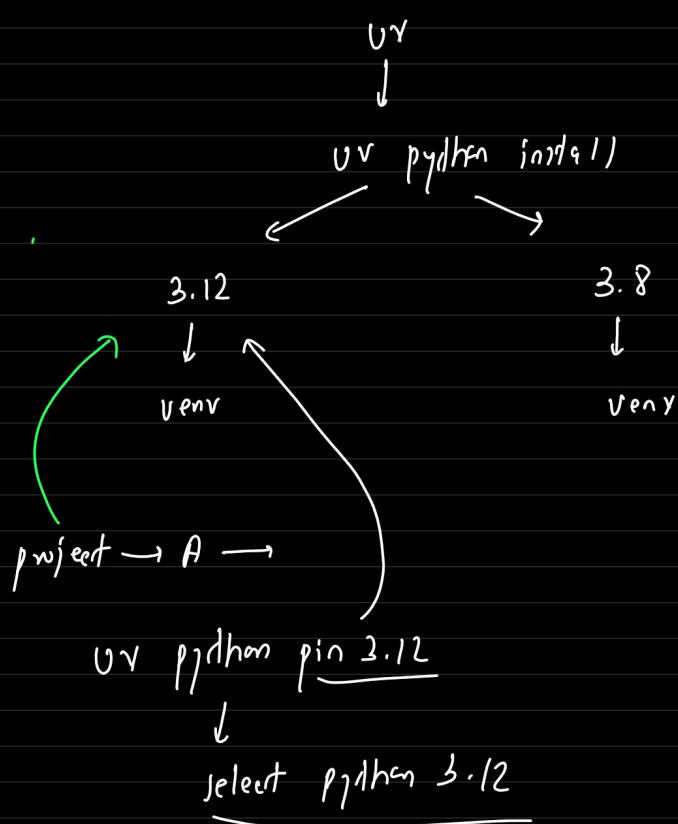
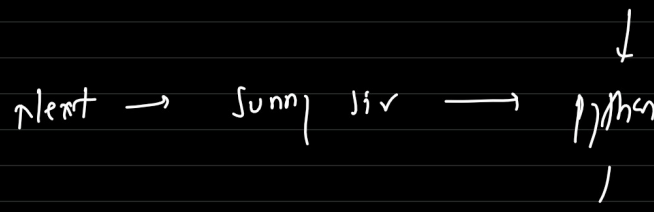
3.6
↓
python → library/module → venv → environment
↓
executable
↑
numpy ← 1 version of numpy at a time
project ← numpy 1.2.1
project ← numpy 1.4.1

python 3.6
↑ project → Lib → numpy 1.2
↓ project → Lib → numpy 1.6
python 3.10

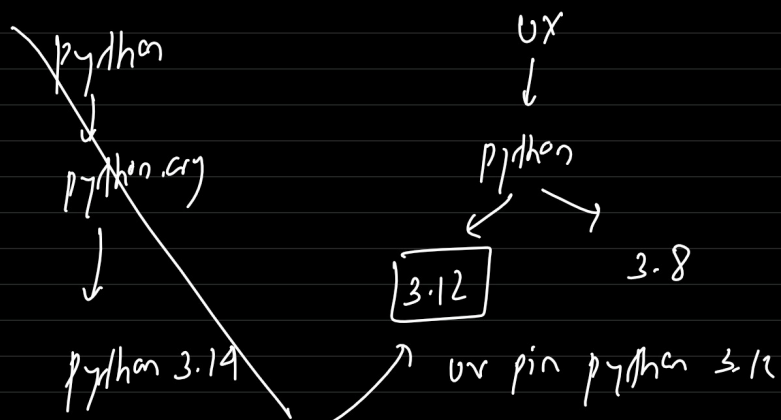
Anaconda → conda
↓
manage different python versions + venv
3.6 → venv
3.7 → venv

VScode: — ✓

uv : —
* watch me do it ✓



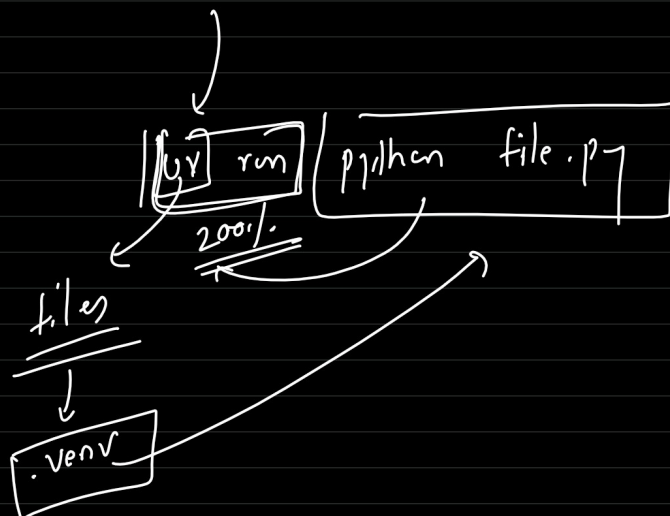
uv add numpy → add library to your venv
uv remove numpy → remove library from venv

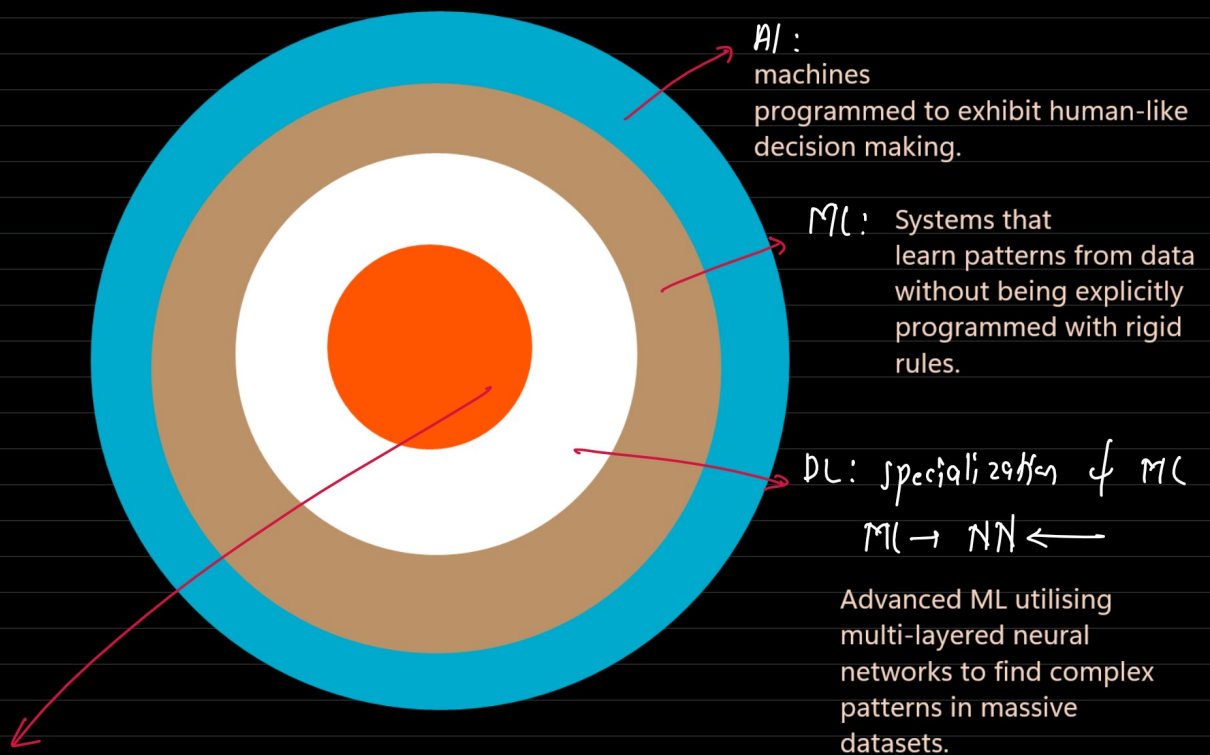


python hello_world.py

windows
.env\Scripts\activate

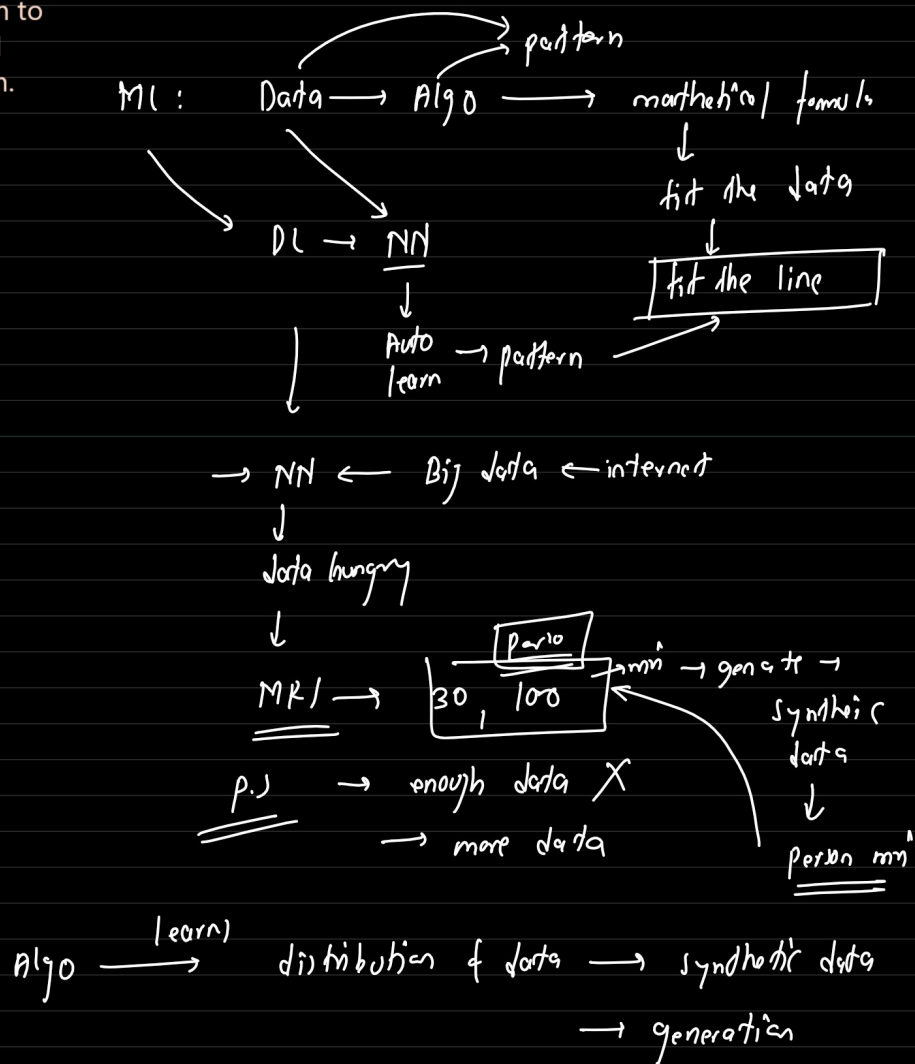
mac/linux
source .env/bin/activate



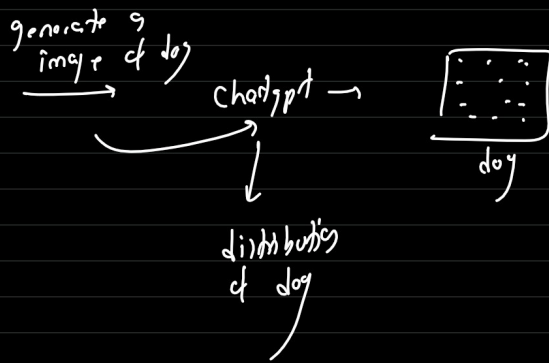


Generative AI:

A specialised branch of Deep Learning that moves beyond classification to create entirely novel content from scratch.

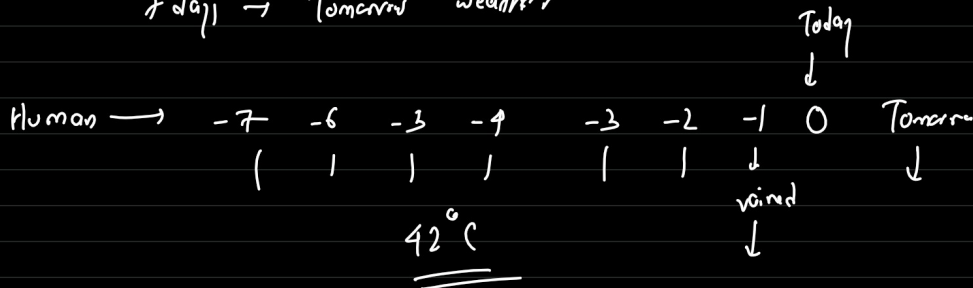


Hi → chatgpt → [] [] [] [] [] []



weather forecast

7 days → Tomorrow weather



if voiced: → fricative hard
un

year

data → ML → partition → fit the line

7 days → ML → Yc
→ NO

generate 7 days of data
containing 7 days of past
date

7 days → GAI → generate → learn the distribution

predict → fit the line

%.
↓
tomorrow weather

10 year

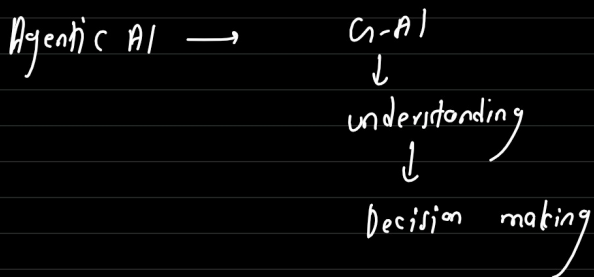
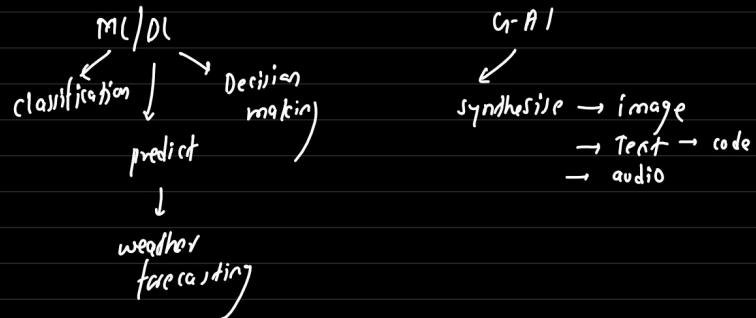
steps

chatgpt → syllable → %.

10 year → analysis pattern →
& where to improve

Traditional $\rightarrow$ Discriminative AI $\rightarrow$ fit the line / learn boundaries

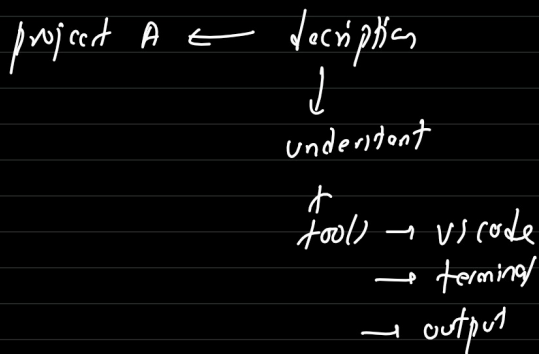
Modern $\rightarrow$ Generative AI $\rightarrow$ learn distribution



System $\rightarrow$ understand
 $\rightarrow$ perform action / act

chatgpt $\rightarrow$ to call
+ my brother
tool $\rightarrow$ call

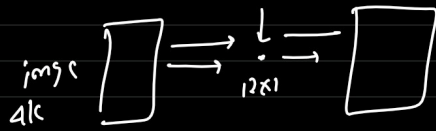
Agent AI



automated

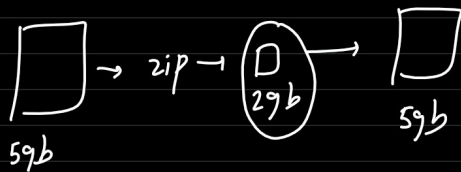
Transformer :- Algo

2013: VAE



2014: GAN

will smdh
eatng spjcdng



data + image

2017: Transformer

algo: understood data from the context

$\rightarrow$ optimized for computation / learning

$\downarrow$
BERT / GPT

$\rightarrow$ speech $\rightarrow$ to $\rightarrow$ text

$\downarrow$
chatgpt
 $\downarrow$
gpt